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CYB102 | Intermediate Cybersecurity

Intermediate Cybersecurity Fall 2024 (@ Section 1 | Tuesdays 5PM - 7PM PDT)

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Including this one, there are only 3 more units left in the individual portion of the course! 🎉

*To keep progressing in our cybersecurity journey, we'll be stepping into some 💪 **more challenging topics**... starting with this lab!*

Take your time and don't worry if it doesn't all make sense right away -- just keep practicing!

Lab #5: Splunkin'

Around

🚩 MACHINE ACCESS CHECK 🚩

To proceed with this unit, please ensure you have access to a **cyb102 Ubuntu VM** as detailed in the [🔗 VM Access Guide](#).

We recommend you begin with **Azure Labs**.

❌ This unit is **incompatible with m-chip Mac / ARM architecture**. If your computer has this processor, you will need to choose a **cloud-based VM option** in the [🔗 VM Access Guide](#).

Overview

In this lab, we'll be exploring our first SIEM tool, [Splunk](#). SIEM (or *Security Information and Event Management*) systems allow us to collect, analyze, and visualize data from various sources to help us identify potential security incidents. Splunk is one of the most popular SIEM tools, and is used by many organizations to help them monitor their networks and systems. This helps them identify potential security incidents, and respond to them quickly and effectively. Let's get [splunking](#)!

If your **Splunk Enterprise demo license** is expiring, you can switch the demo license to a free license by running:

```
sudo /opt/splunk/bin/splunk edit licenser-groups -name "Free" -is_active "1"  
sudo /opt/splunk/bin/splunk restart
```

in the terminal on your VM. In case you are prompted for credentials, these would be the codepath/codepath credentials used to login to Splunk.

Goals

By the end of this lab, you will be able to:

- Understand the Splunk interface
- Build and run basic searches in Splunk
- Create Splunk dashboards, reports, and other visualizations
- Analyze a cybersecurity dataset and identify potential security incidents.

Resources

- [Splunk Documentation](#)
- [Splunk Reference Guide](#)
- [Free Splunk Fundamentals eLearning Courses](#)

Lab Instructions

Required Features

Step 0: Set up the Splunk environment

In this step, you'll RDP into your Ubuntu Machine and access the preconfigured Splunk instance through your browser at <http://127.0.0.1:8000>.

1. RDP (not SSH) into the Ubuntu Box using the provided credentials
2. Open a browser and navigate to <http://127.0.0.1:8000>
3. Log in to Splunk using the username `codepath` and password `codepath`

 **Checkpoint 0:** You should now be logged in to Splunk and able to view the main dashboard.

Step 1: Understand Splunk components and terminology

Welcome to the Splunk lab! Splunk is a **big** data tool.

Q: Why would we want to use Splunk? 🤔 (click for answer)

In brief, Splunk is like **a library full of books**. It's a tool that allows us to store, organize, and analyze large amounts of data. In this lab, we'll be exploring how Splunk works and how it can be used as a SIEM tool to detect and respond to security incidents in a timely manner.

Key Terminology

To organize data effectively, Splunk classifies data into several important fields such as `Index`, `Sourcetype`, `Source`, `Host`, and `Time`. These components work together to categorize and sort data, making it easy to search and analyze. Let's make sure we've got these down as they are how we usually build searches out.

💡 For each term below, guess what you think it might mean... then click to reveal the answer in the Splunk context!

Index

Sourcetype

Source

By separating out these fields it gives us the ability to make our searches more specific for what we're looking for.

- For example, We may want to search books from the same author over various genres much like we might want to search for the same IP address over various server types.

We also break information into two more main categories... WHERE and WHEN:

Host

Time

That was a lot of information! Feel free to reference this section any time you need a refresher on Splunk terminology.

Data Sources

We'll be looking at two different datasets in this lab:

1. Top Video Game Sales:
<https://www.kaggle.com/datasets/thedevastator/global-video-game-sales>
2. An `auth.log` file for a Linux Web Server:

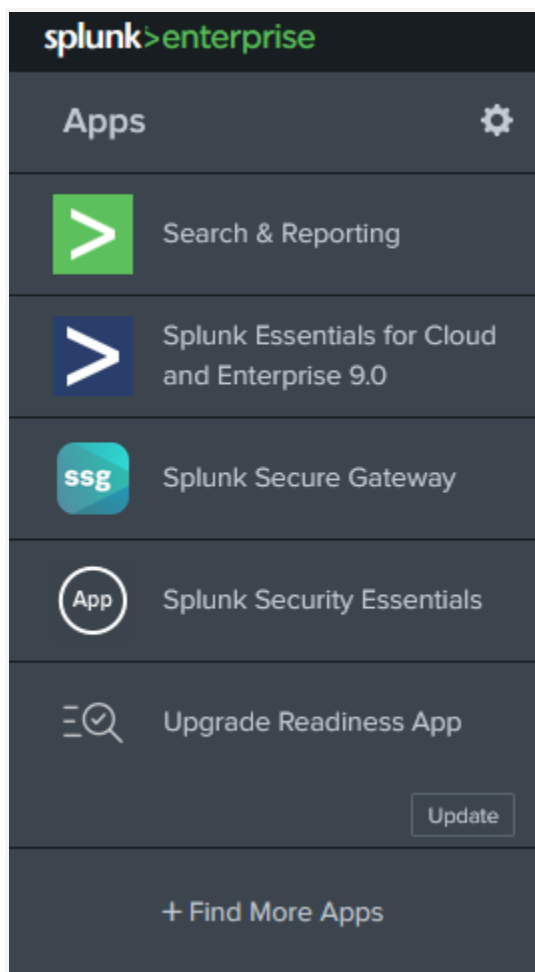
```
66 Apr 29 14:21:20 webserver01 sshd[22]: Failed password for root from 1.3.3.7 port 4321 ssh2
67 Apr 29 14:21:23 webserver01 sshd[22]: Failed password for root from 1.3.3.7 port 4321 ssh2
68 Apr 29 14:22:01 webserver01 sshd[22]: Accepted password for user10 from 70.80.90.100 port 3456 ssh2
69 Apr 29 14:23:10 webserver01 sshd[22]: Failed password for root from 192.168.1.107 port 1234 ssh2
70 Apr 29 14:24:01 webserver01 sshd[22]: Accepted password for user11 from 80.90.100.110 port 4567 ssh2
```

Big data sets are everywhere, and they're extremely useful... if you can find the information you need! Unfortunately, many programs only allow you to search one field at a time, or provide a complicated interface for filtering. Splunk makes it easy to search across multiple fields and quickly find the information you need.

Step 2: Build and run basic searches

In this step, you'll learn how to build and run basic searches using the provided datasets.

Step 2.1: Open the Search & Reporting app



Step 2.2: Our first search

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

New Search Search Bar

index=main host="SalesData"

33,438 events (before 9/16/24 2:30:06.000 PM) No Event Sampling

All Time All time

Events (33,438) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 month per column

Fields to Select **Matching Events**

SELECTED FIELDS
host 1
source 1
sourcetype 1

INTERESTING FIELDS
Critic_Count 100+
Critic_Score 82
Developer 100+
EU_Sales 100+
Genre 12
Global_Sales 100+
Index 1
JP_Sales 100+

i	Time	Event
>	1/11/24 12:33:01.000 PM	Winning Post 8 2016,PSV,2016,Simulation,Tecmo Koei,0,0,0.01,0,0.01,,,,, host = SalesData source = /home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv sourcetype = csv
>	1/11/24 12:33:01.000 PM	Spirits & Spells,GBA,2003,Platform,Wanadoo,0.01,0,0,0.01,,,,, host = SalesData source = /home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv sourcetype = csv
>	1/11/24 12:33:01.000 PM	Haltaka no Psychedelica,PSV,2016,Adventure,Idea Factory,0,0,0.01,0,0.01,,,,, host = SalesData source = /home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv sourcetype = csv
>	1/11/24 12:33:01.000 PM	LMA Manager 2007,X360,2006,Sports,Codemasters,0,0.01,0,0,0.01,,,,, host = SalesData source = /home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv sourcetype = csv
>	1/11/24 12:33:01.000 PM	Samurai Warriors: Sanada Maru,PS3,2016,Action,Tecmo Koei,0,0,0.01,0,0.01,,,,, host = SalesData source = /home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv sourcetype = csv

Step 2.3: Exploring the data

List Format 20 Per Page

i	Time	Event
✓	1/11/24 12:33:01.000 PM	Winning Post 8 2016,PSV,2016,Simulation,Tecmo Koei,0,0,0.01,0,0.01,,,,, Event Actions

Type	Field	Value	Actions
Selected	host	SalesData	✓
	source	/home/codepath/Files/Splunk-5-6-7/Top Video Game sales.csv	✓
	sourcetype	csv	✓
Event	EU_Sales	0	✓
	Genre	Simulation	✓
	Global_Sales	0.01	✓
	JP_Sales	0.01	✓
	NA_Sales	0	✓
	Name	Winning Post 8 2016	✓
	Other_Sales	0	✓
	Platform	PSV	✓

New Search

1 index=main host="SalesData" Genre=Simulation

✓ 1,748 events (before 9/16/24 2:56:57.000 PM) No Event Sampling ▼

Events (1,748) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 20 Per Page ▼

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS # Critic_Count 57 # Critic_Score 56 a Developer 100+ # EU_Sales 90 a Genre 1 # Global_Sales 100+ a index 1 # JP_Sales 70 # linecount 1 # NA_Sales 100+ a Name 100+ # Other_Sales 40 a Platform 23		>	1/11/24 12:33:01.000 PM	Winning Post 8 2016,PSV,2016,Simulation,Tecmo Koei,0,0, host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	Plushees,DS,2008,Simulation,Destineer,0.01,0,0,0.01,, host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	Pony Friends 2,PC,2009,Simulation,Eidos Interactive,0,0, host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	Help Wanted: 50 Wacky Jobs (jp sales),Wii,2008,Simulati host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	National Geographic Panda (JP sales),DS,2008,Simulation host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	Farming 2017 - The Simulation,PS4,2016,Simulation,UIG E host = SalesData source = /home/codepath/Files/Splunk-5-
		>	1/11/24 12:33:01.000 PM	Muv-Luv Alternative,PS3,2012,Simulation,5pb,0,0,0.01,0, host = SalesData source = /home/codepath/Files/Splunk-5-

Step 3: Using stats, count, and creating visualizations

Just looking at the data is great, but the really great stuff comes when we **analyze** and **visualize**.

In this step, you'll learn how to use various stats commands to help create visualizations to better understand the data in the Top Video Game Sales dataset.

Step 3.1: 🔍 Analyze

Step 3.2: A Research Question

New Search

Save As>Create Table ViewClose

1index=main host="SalesData" | stats count by Genre | sort -count

All time

Q

✓ 33,438 events (before 9/16/24 3:14:56.000 PM)No Event SamplingJob|||Smart Mode

EventsPatternsStatistics (12)Visualization

20 Per PageFormatPreview

Genre	count
Action	674
Sports	485
Misc	350
Role-Playing	380
Shooter	264
Adventure	260
Racing	245
Platform	177
Simulation	174
Fighting	165
Strategy	136
Puzzle	116

Step 3.3: 📊 Visualize

●

New Search

1 index=main host="SalesData" | stats count by Genre | sort -count

✓ 33,438 events (before 9/16/24 3:14:56.000 PM) No Event Sampling ▼

Events Patterns Statistics (12) **Visualization**

Column Chart

Format

Trellis

Splunk Visualizations



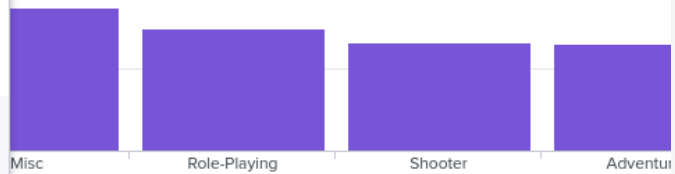
[Find more visualizations](#)

Column Chart

Compare values or fields.

Search Fragment

| stats count by comparison_category



New Search

Save As ▼ Create Table View Close

1 index=main host="SalesData" | stats count by Genre | sort -count | head 5

All time 🔍

✓ 33,438 events (before 9/16/24 3:35:18.000 PM) No Event Sampling ▼

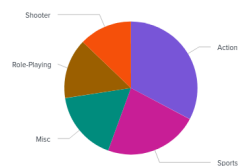
Job ▾ || | ⚙ ⬇ ⬆ Smart Mode ▾

Events Patterns Statistics (5) **Visualization**

Pie Chart

Format

Trellis



Step 4: Using dashboards

In this step, you'll learn how to make and utilize dashboards to view relevant incident information more clearly.

Step 4.1: Creating dashboards

○

Save Panel to New Dashboard



Dashboard Title

Video Game Information

video_game_information

Edit ID

Description

For our video game data

Permissions

Private

How do you want to build your dashboard?

[What's this?](#)

Classic Dashboards

The traditional Splunk dashboard builder

Dashboard Studio

NEW

A new builder to create visually-rich, customizable dashboards

Panel Title

Optional

Visualization Type

Pie Chart

Statistics Table

> Advanced Panel Settings

Cancel

Save to Dashboard

Step 4.2: Editing dashboards

Search
Analytics
Datasets
Reports
Alerts
Dashboards

Edit Dashboard
UI
Source
+ Add Panel
+ Add Input
Dark Theme

Find

New (15)
Events
Statistics Table
Line Chart
Area Chart
Column Chart
Bar Chart
Pie Chart
Scatter Chart
Bubble Chart
Single Value
Radial Gauge
Filler Gauge
Marker Gauge
Cluster Map
Choropleth Map
New from Report (8)
Clone from Dashboard (5)
Add Prebuilt Panel (0)

Video Game Information

For our video game data

Top 5 video game platforms

Top 5 video game platforms

Step 4.3: Try it yourself

Video Game Information

For our video game data

Top 5 video game platforms

Top 5 video game platforms

Top 10 Genres

Genre	count
Action	3378
Sports	2348
Misc	1758
Role-Playing	1500
Shooter	1323
Adventure	1303
Racing	1249
Platform	888
Simulation	874
Fighting	849

Top Game Publishers in 1997

Top Game Publishers in 1997

Step 5: Creating Reports

In this final step, you'll learn how to create reports in Splunk. Reports are useful for summarizing data and presenting it in a consistent format, which can be shared with others or scheduled to be delivered at regular intervals. For this exercise, you'll be switching to a new dataset:

```
index=main host="WebServer01".
```

In the **Search and Reporting App**, set the **Time Range** to **All Time** and enter the following search string:

```
index=main host="WebServer01"
```

This dataset contains logs from a web server, including information about successful and failed login attempts.

Step 5.1: Finding the threat actor

Save As Report [X]

Title:

Description:

Content: ☒ Statistics Table

Time Range Picker: ☒ Yes ☐ No

 Congratulations, you've completed your first Splunk lab! We'll be using Splunk for this unit's project, so make sure to practice these skills.

Stretch Features

Step 6: Automating the process (OPTIONAL)

This step continues where step 5.1 leaves off. If you would like to attempt it:

 **Click to expand instructions**

Edit Schedule



 Scheduling this report results in removal of the time picker from the report display.

Report	Failed IPs
Schedule Report	☒
	Learn More
Schedule	Run every week ▼
On	Monday ▼ at 6:00 ▼
Time Range	All time ▶
Schedule Priority ?	Default ▼
Schedule Window ?	No window ▼
Trigger Actions	
	+ Add Actions ▼

Cancel

Save

Step 7: Do Your Own Splunk

If you have your own dataset that you'd like to analyze in Splunk, you can upload it to Splunk and begin exploring the data. Here are some instructions on how to upload a CSV file to Splunk:

First, log in to your Splunk account and navigate to the Splunk search bar.

Click on the **Add Data** button in the top left corner of the screen.
Select the **Upload** option from the menu.
Click the **Select File** button to browse your computer for the CSV file that you want to upload.
Once you have selected the file, click the **Next** button.
On the **Set Sourcetype** page, select **From list of sourcetypes** and choose **CSV** from the list.
Click the **Next** button to proceed.
On the **Set Timestamp** page, select **No timestamp** and click **Review**.
Review the settings to make sure they are correct, then click **Submit** to upload the file.

Once the upload is complete, you should see a message indicating that the data has been successfully added.

You can now search and analyze the data in Splunk by typing your search query in the search bar.

That's it! Your CSV file should now be uploaded to Splunk and ready for analysis. Keep in mind that depending on the size of your file, it may take some time to upload and process the data.

Share with a neighbor: Find a partner and take turns explaining what dataset you chose, how you analyzed the data, and anything else you found interesting!

You've completed your lab AND stretch goals! 🚀

💡 **TIP:** Remember to come back and reference this lab when you need to do similar things in your project!

Extra Practice

- If you want to install Splunk on your own systems, you can follow the guide [here](#).